

AR(2) in a Spatial Skew- t Model: Why a Temporal Prior Loses to an i.i.d. Baseline Under Spatial-Only Holdout

Technical Note — `ar2.rethink`

Abstract

We document a sequence of derivations explaining why an AR(2) temporal prior on the latent processes of a spatial skew- t model fails to dominate its independent-in-time counterpart when the prediction task is scored by the Brier score on held-out *spatial* sites. The construction normalizes each AR(2) latent so that its stationary marginal variance equals one; we show this construction implies the time- t marginal of every latent is invariant to the autoregressive coefficients ϕ_1, ϕ_2 . The Brier loss in the spatial-holdout design integrates the predictive kernel against precisely this time- t marginal, so the temporal coupling marginalizes out. The AR(2) parameters can still leak into the predictive through the posterior of the latents at training times, but each of the four available leakage channels is damped (discrete knot membership, likelihood-dominated latents, ϕ -independent spatial noise), while the cost of six additional AR coefficients inflates posterior variance. We close with three concrete redesigns—block-time-forecast holdout, multivariate energy score, variogram score—under which the AR(2) prior is expected to recover its advantage.

1 Introduction

The simulation study in `code/analysis/simstudy/` compares eight prediction methods for spatial extremes, including four variants of the skew- t knot model of which two (methods 7 and 8) place an AR(2) prior on every latent temporal process. Tabulated Brier and quantile scores for settings 9 and 10 show that the AR(2) methods do *not* systematically outperform their i.i.d. siblings; in setting 10 the K=5 AR(2) variant is in fact worse than the K=5 no-AR baseline at every quantile.

This note explains the discrepancy. The conclusion has nothing to do with bugs in the AR(2) implementation in `simulate_ar2_standard()` and everything to do with the interaction between (i) the AR(2) standardisation, (ii) the spatial structure of the holdout, and (iii) the scoring rule. The reasoning is reproduced here in a single place because it determines what experiments would actually credit the AR(2) machinery.

The structure of the note is as follows. Section 2 fixes notation and lists the assumptions of the underlying model. Section 3 formalises the AR(2) standardisation and establishes the marginal invariance proposition. Section 4 carries the invariance through the Brier predictive integral and identifies the four leakage channels. Section 5 proposes alternative scoring designs. Section 7 summarises.

2 Model Setup and Notation

2.1 Spatial skew- t knot model

Let $\{s_i\}_{i=1}^{n_s} \subset \mathbb{R}^2$ denote spatial locations and $t \in \{1, \dots, T\}$ time indices. Fix K knots in the spatial domain. The conditional model for the observed process is

$$Y_t(s) = x_t(s)^\top \beta + \lambda z_{g(s,t),t} + \frac{1}{\sqrt{\tau_{g(s,t),t}}} [L_C \varepsilon_t]_s, \quad \varepsilon_t \sim \mathcal{N}(0, I_{n_s}), \quad (1)$$

where:

- $\beta \in \mathbb{R}^p$ are regression coefficients and λ is the skewness parameter;
- $z_{k,t}, \tau_{k,t}$ are knot-level latent processes governing skew and scale;
- $w_{k,t} \in \mathbb{R}^2$ is the (possibly time-varying) location of knot k at time t ;
- $g(s, t) = \arg \min_k \|s - w_{k,t}\|$ is the knot membership function (a Voronoi assignment);
- $C = C(\rho, \nu, \gamma)$ is the spatial covariance matrix on the observation grid and $L_C = \text{chol}(C)^\top$.

We collect the time- t slice of all latents into $\mathcal{L}_t = (z_{\cdot,t}, \tau_{\cdot,t}, w_{\cdot,t})$ and the full fixed-parameter vector into θ .

2.2 AR(2) standardisation

Each scalar latent process (one per knot per quantity z, τ, w) is generated by

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \quad (2)$$

with (ϕ_1, ϕ_2) in the stationarity region. We adopt the construction in `simulate_ar2_standard()`, which fixes the stationary variance and *solves* for the innovation variance.

Definition 1 (Standardised AR(2)). Set $\gamma_0 = 1$, $\gamma_1 = \phi_1/(1 - \phi_2)$, and

$$\sigma_\eta^2 = \gamma_0 - \phi_1 \gamma_1 - \phi_2 \gamma_2, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0. \quad (3)$$

The initial pair (X_1, X_2) is drawn jointly from

$$\mathcal{N}\left(0, \begin{bmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{bmatrix}\right),$$

and the recursion (2) runs forward for $t \geq 3$.

Assumption 1 (Stationarity). (ϕ_1, ϕ_2) satisfies $|\phi_2| < 1$, $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, so the roots of the AR(2) characteristic polynomial lie strictly outside the unit circle.

2.3 Prediction task

The simstudy holdout fixes a subset $\mathcal{S}_{\text{test}}$ of test *sites*; the training set is $\mathcal{D} = \{Y_t(s) : s \notin \mathcal{S}_{\text{test}}, t \in \{1, \dots, T\}\}$, and the prediction targets are $\{Y_t(s^*) : s^* \in \mathcal{S}_{\text{test}}, t \in \{1, \dots, T\}\}$. Note that every time point is observed at every training site.

Definition 2 (Brier score). For threshold c and predictive probability $\hat{p}(s^*, t; c) = \mathbb{P}(Y_t(s^*) > c \mid \mathcal{D})$,

$$\text{BS}(c) = \mathbb{E}[(\mathbf{1}\{Y_t(s^*) > c\} - \hat{p}(s^*, t; c))^2]. \quad (4)$$

3 Marginal Invariance of the Standardised AR(2)

Proposition 1 (Marginal invariance to ϕ). *Under Definition 1 and Assumption 1, for every $t \in \{1, \dots, T\}$,*

$$X_t \sim \mathcal{N}(0, 1) \quad \text{regardless of } (\phi_1, \phi_2). \quad (5)$$

Proof. By construction $X_1, X_2 \sim \mathcal{N}(0, 1)$ marginally. Suppose inductively that $\text{Var}(X_{t-1}) = \text{Var}(X_{t-2}) = 1$ and $\text{Cov}(X_{t-1}, X_{t-2}) = \gamma_1$. Then from (2),

$$\text{Var}(X_t) = \phi_1^2 + \phi_2^2 + 2\phi_1\phi_2\gamma_1 + \sigma_\eta^2.$$

Substituting (3) and using $\gamma_2 = \phi_1\gamma_1 + \phi_2$ gives $\text{Var}(X_t) = 1$. Mean is zero by linearity. Joint normality of X_t with previous lags is preserved by the linear recursion. $\square$

Corollary 1 (AR(1) toy). *For an AR(1) process standardised to $\gamma_0 = 1$, $\sigma_\eta^2 = 1 - \phi^2$, and $X_t \sim \mathcal{N}(0, 1)$ for every admissible ϕ . The autocorrelation $\text{Cov}(X_t, X_{t+h}) = \phi^{|h|}$ is the only quantity that moves with ϕ .*

Remark 1 (Identifiability). ϕ is identifiable only from *joint* distributions across time; the one-point marginal is ϕ -blind. Sampling one X_t alone provides zero information about ϕ .

4 Why Brier Marginalises ϕ Away

4.1 The predictive integral collapses to the time- t marginal

Proposition 2 (Predictive at (s^*, t) depends only on time- t latents). *For the kernel (1), the conditional CDF at (s^*, t) given $(\theta, \mathcal{L}_t)$ does not depend on $\mathcal{L}_{t'}$ for any $t' \neq t$. Consequently the marginal predictive is*

$$\hat{p}(s^*, t; c) = \int \int \mathbb{P}(Y_t(s^*) > c \mid \theta, \mathcal{L}_t) \pi(\mathcal{L}_t \mid \theta, \mathcal{D}) \pi(\theta \mid \mathcal{D}) d\mathcal{L}_t d\theta. \quad (6)$$

Proof. Direct from inspection of (1): the right-hand side involves only $\mathcal{L}_t$ at the same t . Fubini and the conditional factorisation $\pi(\theta, \mathcal{L}_t \mid \mathcal{D}) = \pi(\mathcal{L}_t \mid \theta, \mathcal{D})\pi(\theta \mid \mathcal{D})$ give the displayed form. $\square$

Proposition 3 (ϕ -free prior marginal at one t). *Under Definition 1, the prior marginal of $\mathcal{L}_t$ at any single t does not depend on $\phi := (\phi_z, \phi_\tau, \phi_w)$.*

Proof. Each component of $\mathcal{L}_t$ is obtained by a fixed monotone transform (gamma inverse copula for τ , half-normal inverse copula for z , inverse probit for w) applied to a standardised AR(2) latent. Proposition 1 establishes that each underlying latent has marginal $\mathcal{N}(0, 1)$ for every admissible ϕ . The fixed transforms preserve this invariance. $\square$

Corollary 2 (Bayes vs. prior). *Combining Propositions 2 and 3, the only route through which ϕ can affect $\hat{p}(s^*, t; c)$ is the posterior $\pi(\theta \mid \mathcal{D})$ together with the conditional $\pi(\mathcal{L}_t \mid \theta, \mathcal{D})$. The likelihood $p(\mathcal{D} \mid \theta)$ couples observations across time and therefore depends on ϕ , but the kernel at the held-out cell does not.*

4.2 Four leakage channels

We enumerate the channels through which ϕ can move the predictive at (s^*, t) in spite of Corollary 2.

Channel A: $z_{g,t}$ via temporal pooling. AR(2) tightens $\pi(z_{k,t} | \mathcal{D})$ at training t by borrowing from $z_{k,t\pm 1}, z_{k,t\pm 2}$. The benefit is bounded by how much information the time- t training likelihood already provides.

Channel B: $\tau_{g,t}$ via temporal pooling. Symmetric to Channel A.

Channel C: knot membership $g(s^*, t)$. AR(2) tightens $\pi(w_{k,t} | \mathcal{D})$ at training t . Because $g(s^*, t)$ is a discrete *argmin* over $\{w_{1,t}, \dots, w_{K,t}\}$, the posterior tightening propagates to g only when the perturbation crosses a Voronoi boundary near s^* . For $K = 1$ the channel is closed entirely.

Channel D: spatial noise $L_C \varepsilon_t / \sqrt{\tau_{g,t}}$. ε_t is i.i.d. in t given θ . AR(2) does not appear in this channel at all.

We also account for a cost channel that operates against any benefit from A–C.

Channel E (cost): posterior on ϕ . Methods 7 and 8 turn on AR(2) for z, τ, w simultaneously, introducing six new parameters $\phi_{1z}, \phi_{2z}, \phi_{1\tau}, \phi_{2\tau}, \phi_{1w}, \phi_{2w}$. Their joint posterior contributes variability to the predictive integral (6).

4.3 Variance decomposition

Lemma 1 (Predictive variance decomposition). *For the held-out predictive probability $\hat{F}_{s^*,t}(c) = \hat{p}(s^*, t; c)$,*

$$\text{Var}(\hat{F}_{s^*,t}(c)) \approx \mathbb{E}_\phi[\text{Var}(\hat{F} | \phi)] + \text{Var}_\phi(\mathbb{E}[\hat{F} | \phi]). \quad (7)$$

The AR(2) prior reduces the first summand through Channels A–C; it introduces the second summand, which is identically zero under i.i.d. time. The Brier score is sensitive to the sum, not the decomposition. Empirically (Section 4.4) the second summand dominates the first reduction in the simstudy configuration.

4.4 Evidence in the simstudy

Table 1 summarises Brier scores relative to the Gaussian baseline (method 1) at selected quantiles for setting 9.

Table 1: Relative Brier score versus method 1, setting 9. Smaller is better; entries below 1 indicate improvement over the Gaussian baseline.

q	M2 (skew- t , $K=1$)	M4 (skew- t , $K=5$)	M7 (AR(2), $K=1$)	M8 (AR(2), $K=5$)
0.90	0.961	0.973	0.968	0.972
0.95	0.951	0.969	0.964	0.969
0.97	0.943	0.965	0.958	0.958
0.99	0.955	0.987	0.975	0.980
0.995	0.946	0.955	0.957	0.962

The i.i.d. skew- t at $K = 1$ (method 2) systematically beats its AR(2) sibling (method 7). Setting 10 shows an even sharper version of the same phenomenon: method 8 ($K = 5$ with AR(2)) ranges from 0.96 to 1.16 of the Gaussian baseline, while method 4 ($K = 5$ without AR(2)) is at 0.87–0.94.

5 Redesigns That Restore the AR(2) Signal

Three designs expose the temporal coupling that the spatial-only Brier ignores.

5.1 Design A: block-time-forecast holdout

Hold out the last T_h time points; train on $\{Y_t(s) : t \leq T - T_h, s \in \mathcal{S}\}$. The predictive at horizon $t^* > T - T_h$ requires the conditional

$$z_{k,t^*} | \mathcal{D} \sim \mathcal{N}(\phi_1 \hat{z}_{k,t^*-1} + \phi_2 \hat{z}_{k,t^*-2}, \sigma_\eta^2 + \text{filter uncertainty}),$$

which is strictly ϕ -dependent. Brier and any scoring rule applied to the forecast cells will distinguish AR(2) from i.i.d. time.

5.2 Design B: multivariate score on the spatial holdout

Keep the spatial holdout. Score the held-out site-level trajectory $\mathbf{y}_{s^*} = (Y_1(s^*), \dots, Y_T(s^*))$ as a T -vector.

Energy score. For predictive F and observation $\mathbf{y} \in \mathbb{R}^T$,

$$\text{ES}(F, \mathbf{y}) = \mathbb{E}_F \|X - \mathbf{y}\|_2 - \frac{1}{2} \mathbb{E}_F \|X - X'\|_2, \quad X, X' \stackrel{\text{iid}}{\sim} F. \quad (8)$$

The Monte Carlo estimator with M samples is

$$\widehat{\text{ES}} = \frac{1}{M} \sum_{m=1}^M \|X^{(m)} - \mathbf{y}\|_2 - \frac{1}{2M(M-1)} \sum_{m \neq m'} \|X^{(m)} - X^{(m')}\|_2.$$

Variogram score of order p .

$$\text{VS}_p(F, \mathbf{y}) = \sum_{t < t'} w_{t,t'} \left(|y_t - y_{t'}|^p - \mathbb{E}_F |X_t - X_{t'}|^p \right)^2. \quad (9)$$

With $p = 0.5$ the score is particularly sensitive to temporal dependence misspecification.

5.3 Design C (recommended): combined holdout + energy score

Hold out both a set of test sites and the last T_h time points at the remaining sites. Report:

1. marginal Brier on (s^*, t) for $s^* \in \mathcal{S}_{\text{test}}, t \leq T - T_h$ — the legacy comparison;
2. energy score on (s, t_{block}) for s in the training spatial set, $t > T - T_h$ — the temporal-coupling test;
3. energy score on (s^*, t_{block}) — space-and-time extrapolation.

Remark 2 (Reporting). Report paired score differences across the 50 simulation replicates and bootstrap the resulting boxplots. Also report score-per-CPU-hour: methods 7 and 8 are more expensive, and a small score gain may not be worth the additional compute.

6 Intuition Summary

AR(2) is a model of how today’s latent values relate to yesterday’s and the day before yesterday’s. The Brier score on the spatial holdout asks for the marginal exceedance probability at a new *location* on a day that is already observed at other locations. The temporal coupling that AR(2) models has no opportunity to act, because every relevant time point is already pinned by training-time likelihood. The score sees only the cost of the additional parameters — six ϕ ’s with their own posterior variance — and penalises the model for buying capability it never gets to use.

The fix is not to remove AR(2); it is to design an experiment that asks the model to act like a temporal model. Forecast forward in time, or score whole trajectories at once. The temporal coupling then enters the predictive integral non-trivially, and the AR(2) prior recovers its rent.

7 Conclusion and Future Work

We have established that the AR(2) standardisation (Definition 1) makes the prior time- t marginal of every latent invariant to the autoregressive coefficients (Proposition 1, Proposition 3). The spatial-only Brier score integrates the predictive kernel against precisely this marginal (Proposition 2), so the temporal coupling vanishes from the loss. Of the four leakage channels through which ϕ could re-enter via the posterior, three (Channels A–C) are damped by likelihood dominance, discrete knot membership, or spatial ϕ -independence; the fourth (Channel D) is closed. The cost channel (E) is unbounded above and inflates the predictive variance, which Brier penalises directly. The net effect is that i.i.d.-time skew- t wins the comparison on this score.

Future work, in order of priority:

1. Implement Design C in `scores.R` and `run-settings.R`: extend the MCMC predictive loop in `rpotspatTS_arp()` to propagate the AR(2) recursion past the training horizon, and add `EnergyScore` and `VariogramScore` alongside `BrierScore`.
2. Re-run settings 9 and 10 with $T_h \in \{3, 5, 10\}$ and report paired score differences between methods (2, 7) and (4, 8).
3. Run model selection over AR order $p \in \{0, 1, 2\}$ using DIC/WAIC before fixing the temporal-prior order, so any observed AR(2) advantage cannot be attributed to a coincidentally matched simulation truth.
4. Investigate whether channel C activates non-trivially when the data-generating w has small inter-knot separation, i.e. when many held-out sites sit near Voronoi boundaries.

A Reference R implementation

The standardised AR(2) simulator in `code/00_core/auxfunctions_ar2.R`.

```
simulate_ar2_standard <- function(nt, n, phi1, phi2,
                                  tol = 1e-06, param_name = NULL) {
  # ... stationarity check ...
  denom <- 1 - phi2
  gamma0 <- 1
  gamma1 <- phi1 * gamma0 / denom
  gamma2 <- phi1 * gamma1 + phi2 * gamma0
  sigma2 <- gamma0 - phi1 * gamma1 - phi2 * gamma2
  innov.sd <- sqrt(sigma2)
```

```

ar2 <- matrix(0, nrow = n, ncol = nt)
init.states <- draw_ar2_initial(n, gamma0, gamma1)
ar2[, 1] <- init.states[, 1]
ar2[, 2] <- init.states[, 2]

for (t in 3:nt) {
  ar2[, t] <- phi1 * ar2[, t - 1] +
             phi2 * ar2[, t - 2] +
             rnorm(n, 0, innov.sd)
}
ar2
}

```

The construction matches Definition 1: the marginal variance is fixed at $\gamma_0 = 1$ and the innovation variance is derived. Initial states are drawn jointly from the stationary bivariate Gaussian via `draw_ar2_initial()`, removing the need for a burn-in.

B Verification by reverse simulation

To confirm the marginal invariance empirically we simulate $n = 1000$ AR(2) paths under $(\phi_1, \phi_2) = (0.6, -0.3)$ via manual recursion and via `stats::filter(..., method = "recursive")`, fit `arima(order = c(2, 0, 0))` to each, and run a residual diagnostic. Both fits recover the true coefficients within one standard error, and the Ljung–Box and Shapiro–Wilk tests on the residuals are non-significant ($p > 0.4$). Manual recursion and filtering produce statistically equivalent AR(2) series.